Parth Sharma

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AI Engineer

AI Engineer with 4 years of experience building and deploying production-grade AI/ML and web applications at enterprise scale. Specialized in Agentic AI, RAG, and multimodal platforms using Python, Golang, TypeScript, with cloud-native deployments across AWS, Azure, and GCP. Co-author of 7 IEEE publications and inventor on a US patent application.

SKILLS

Programming Languages: Golang, Python, TypeScript

Frameworks & APIs: Celery, FastAPI, Gin-Gonic, LangChain, Langfuse, LangGraph, Next.js, OpenTelemetry

AI & ML: LLMs, Generative AI, Agentic AI, RAG, Computer Vision, Reinforcement Learning, Vector DBs, Neural Networks

Cloud & DevOps: AWS, Azure, Caddy, Docker, GCP, GitHub Actions, GitLab CI/ CD, Kubernetes, NGINX

PATENTS

Dynamic Retrieval & Indexing Framework Technology (DRIFT) 05/2025

- *US Patent Application No. 19/200080*, Filed: May 6, 2025
- Developed a novel semantic-fixed length segmentation mechanism (DRIFT) to preserve contextual integrity in large-scale indexing.

WORK EXPERIENCE

Genpact **Bengaluru, Karnataka, India**

Assistant Manager | AI Practice • Full-time **03/2024 - Present**

Intelligent Observability & Debugging Platform

Gin-Gonic, Next.js, Langfuse, GitHub MCPs, OpenTelemetry, TRPCs

- Led development of an Agentic AI-powered observability and debugging platform for Python applications, enabling **zero effort telemetry injection** and contributing to **Genpact becoming an AWS Agent Marketplace launch partner**.
- Built a **high-performance asynchronous Golang service (Gin-Gonic)** to fetch and refactor code from Git branches via GitHub MCP servers, processing **1k+ files per run** with **concurrent execution**, achieving **~3× faster refactoring throughput** while ensuring race-free execution using Go channels and wait groups.
- Designed and integrated a **Next.js + Golang agentic AI architecture** with Langfuse and GitHub MCP tools to analyse tracebacks, auto-generate fixes, and create bug-fix branches, reducing **mean time to resolution (MTTR) by ~40%** for recurring code issues.
- Contributed to Langfuse open-source repository; resolved 2 critical bugs related to python trace ingestion, improving stability for ~1k+ active users.

Multi-Modal Agentic Vision System

FastAPI, Next.js, Celery, Redis, Azure, WebSockets

- Engineered an **Agentic AI vision system** that processes live video feeds via **FastAPI** and **Next.js**, enabling real-time incident detection and semantic video search.
- Built a **high-concurrency ingestion pipeline** with Redis-Celery and MoviePy to decode, frame, and index video at **5-second intervals**, sustaining **20+ parallel streams** and processing **10k+ frames per hour** with synchronized temporal metadata.
- Developed a multi-stage retrieval architecture that utilizes **vector search** to identify relevant time-stamps, followed by a **Vision Language Model (VLM)** to cross-verify visual incidents against agent-driven queries.
- Implemented **real-time server-push notifications** and live status updates using WebSocket, achieving **near–real-time alerting (<10s)** for critical incidents.

Data Scientist | AI Practice • Full-time **08/2022 - 03/2024**

Knowledge Assist | Multi-Tenant Application

React, FastAPI, Celery, Docker, GitHub Actions, Azure

- Architected and deployed a production-grade multi-tenant knowledge assist platform using **React, FastAPI, and Celery**, delivering a scalable knowledge solution sold to **20+ enterprise clients**.
- Designed and implemented **SPARKS (Semantic and Page-wise Augment Knowledge Retrieval System)**, a semantic segmentation–based search engine that preserved document context and narrative flow, improving **retrieval relevance by ~25– 30%** over traditional fixed-size chunking approaches *(USPTO pending)*.
- Engineered a high-performance asynchronous processing pipeline using **Celery and Redis**, **reducing document indexing time by 20% and cloud infrastructure costs by 35%** streamlining the application architecture

CCS - STRATEGY SOLUTIONS NEW DEHLI

New Delhi, Delhi, India

Intern **05/2020 - 07/2020**

- Created a 3D Model of an Automated Plate Mill Production Factory using FlexSim Software
- Assisted in formulation of schedules for machine operations and overall optimization of the process.

PROJECTS

Centre of Excellence (CoE)

Multi-Tenant Enterprise CMS (Onboarding Website, Staging Site)

- Architected and deployed two high-traffic, **multi-tenant React/Next.js** platforms, optimizing tenant onboarding and data isolation using **Golang** backends **reducing 60% costs compared to previous implementation**.
- Integrated complex **GraphQL** based third-party services including Shopify Hydrogen and Sports Radar APIs, **currently serving 15+ global clients** (Australia).

Multi-Tenant AI Dealer Dashboard

- Designed and developed a **multi-tenant AI-powered dealer dashboard** for leading car dealership using **Next.js**, enabling centralized tracking of sales, service, and operational KPIs across **20+ dealer accounts**.
- Integrated **Generative AI** to generate automated executive summaries, daily performance insights, and exception alerts, reducing manual reporting effort by **~40%**.

PUBLICATIONS

Institute of Electrical and Electronics Engineers (IEEE)

- Dynamic Thresholding in Document Indexing: An Adaptive Scoring Approach for Retrieval Optimization
- Semantic Page-wise Augmented Knowledge Retrieval System: SPARKS
- Dynamic Retriever Selection in RAG Systems: An RL Approach to User-Centric NLP

Additional publications upon request

EDUCATION

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering - BE in Manufacturing Engineering Pilani • 08/2018 - 07/2022

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals AZ-900

05/2024 - Present

Microsoft